From AI Safety Regulation to Sociotechnical Evaluation

From model capability to evidence about real-world impact

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I think it is safe to say

We all want ‘good’ AI

Slides adapted from Anders Skaarup Johansen (18 Oct 2026)

If a model passes our safety benchmarks, can we conclude that it is safe in use?

Today:from benchmark scores to sociotechnical evaluation

1. **AI Regulation frameworks**
   Governance and regulations' broad expectations
2. **Sociotechnical Eval of AI safety**
   Three-layer evaluation framework
3. **AI Safety frameworks and system cards**
   Lab-specific safety evidence and documentation
4. **Beyond benchmark scores**

Governance broad goals; evaluation must supply evidence

Framework	Type	Main emphasis
EU AI Act	EU	Risk-based obligations for general-purpose models, systemic risks
ISO/IEC 42001 and 42005	UK	AI management and system impact assessment
NIST AI RMF	US	Identifying, measuring, and managing AI risks

Source: EU AI Act, Arts. 27 and 55 [3]; ISO/IEC 42005 [5]; Council of Europe [4]; NIST [8]



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"High-risk AI systems shall be designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately. An appropriate type and degree of transparency shall be ensured with a view to achieving compliance with the relevant obligations of the provider and deployer set out in Section 3."
- AI-Act: Article 13.1

"its intended purpose"
- AI-Act: Article 13.3 (i)

"the level of accuracy, including its metrics, robustness and cybersecurity referred to in Article 15 against which the high-risk AI system has been tested and validated and which can be expected, and any known and foreseeable circumstances that may have an impact on that expected level of accuracy, robustness and cybersecurity;"
- AI-Act: Article 13.3 (ii)

"where applicable, the technical capabilities and characteristics of the high-risk AI system to provide information that is relevant to explain its output"
- AI-Act: Article 13.3 (iv)

"when appropriate, specifications for the input data, or any other relevant information in terms of the training, validation and testing data sets used, taking into account the intended purpose of the high-risk AI system **- AI-Act: Article 13.3 (vi)**

"where applicable, information to enable deployers to interpret the output of the high-risk AI system and use it appropriately"
- AI-Act: Article 13.3 (vii)

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The operationalization gap

The challenge

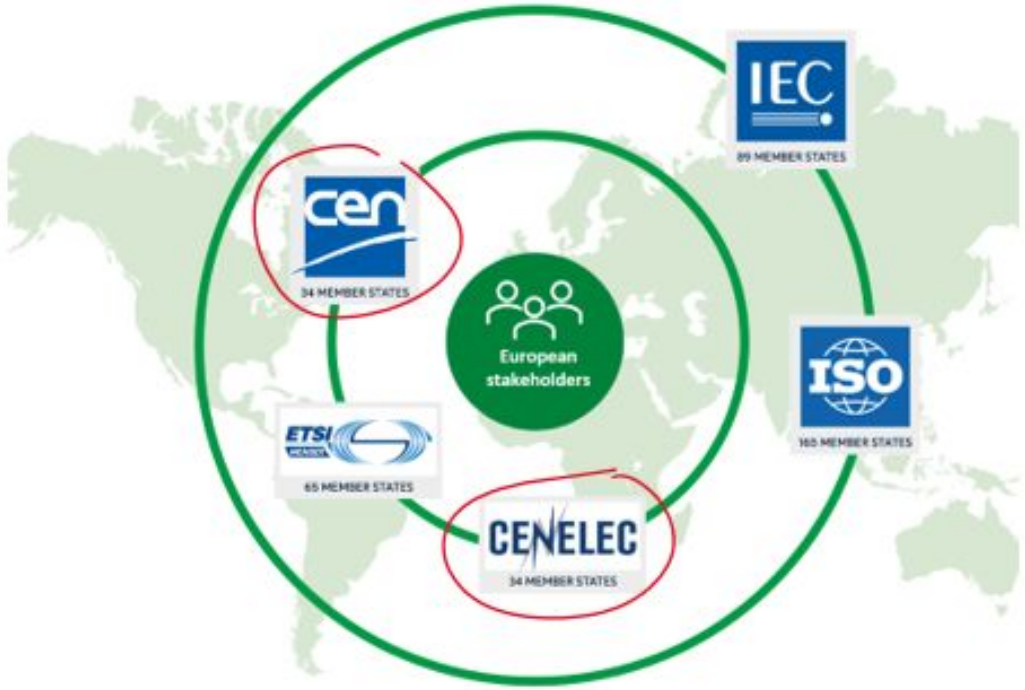
Guidance exists, but each evaluation still needs context-specific choices about constructs, methods, baselines, and decision thresholds.

Governance goal: preserve human agency	**Evaluation choice still needed**
Meaningful control	Which behavior demonstrates control?
Recognition of limitations	Does a disclaimer improve understanding?
Ability to challenge advice	Can users detect, reject, and correct errors?
Evidence informs deployment	Which failure rate triggers redesign or restricted use?

Source: NIST AI RMF Playbook [8]; Council of Europe HUDERIA [9]

Regulation > Standard > Benchmark

The European Commission asked for standardized technical specifications



Slides adapted from Anders Skaarup Johansen (18 Oct 2026)

The European Commission asked for standardized technical specifications



Deadline August 2025?

CEN CENELEC

REGULATION (EU) No 1025/2012
defines a harmonized standard as:
'harmonized standard' means a European standard adopted on the basis of a request made by the Commission for the application of Union harmonization legislation.

https://www.nsai.ie/images/uploads/general/NSAI_AI_report_digital.pdf

Slides adapted from Anders Skaarup Johansen (18 Oct 2026)



Slides adapted from Anders Skaarup Johansen (18 Oct 2026)

The organization shall select one or more metrics for the evaluation of the task. For each selected metric, the following items shall be documented:

- **Name and definition of the metric.** The name of the metric and its mathematical formula with all parameters explicitly defined.
- **Applicability.** The justification for selecting this metric for this task, specifying which aspect of output quality the metric is intended to measure, and why this aspect is relevant to the intended purpose of the system.
- **Limitations.** The known limitations of the metric, including any aspects of output quality that the metric does not capture, any known biases or sensitivities (e.g. sensitivity to class imbalance, depth range, or outlier values), and any conditions under which the metric may produce misleading results.
- **Application details.** This notably includes aspects to take care of when applying that metric to that particular task. This may include any pre-processing, alignment, or filtering applied to the system before metric computation.
- **Implementation details.** This notably includes technical aspects to take care of when implementing the metric for that particular task (e.g. data encoding, scaling conventions, floating-point precision).
- **Calibration.** Where available, the typical range of metric values observed when the metric is applied to this task. This information is informative. Where calibration data is not available, the organization shall state that no calibration reference has been established and shall justify why evaluation remains valid without it.
- **Interpretation.** Indication of which aspects of output quality the metric is measuring or focusing on (e.g. what a given value indicates about the behaviour of the system, what aspects of performance it reflects).

Slides adapted from Anders Skaarup Johansen (18 Oct 2026)

Sorter forslag efter Relevance

DSF/prEN ISO/IEC 23282 - Kunstig intelligens (AI) – Metoder til evaluering af sprogteknologi (NLP)

Emneområder: S-855 Kunstig Intelligens
Udvalgsnummer: S-855 (Kunstig Intelligens)
Kilde: CENCLC
Slutdato: 11-11-2026

Detaljer på forslag

Se mere

DSF/ISO/IEC DIS 23282 - Kunstig intelligens (AI) – Metoder til evaluering af sprogteknologi (NLP)

Emneområder: S-855 Kunstig Intelligens
Udvalgsnummer: S-855 (Kunstig Intelligens)
Kilde: ISO
Slutdato: 01-11-2026

Detaljer på forslag

Se mere

6.2.7.2 BLEU

6.2.7.2.1 The BLEU score measures the extent to which a candidate sequence (e.g. a sentence) in text matches the form and content of a given reference sequence (or multiple references), accounting for terminology, phrasing, and the possibility of multiple equivalent phrasings for the same sequence.

NOTE Known limitations of BLEU include semantic insensitivity, lexical overlap bias and limited correlation with human ratings, among other empirically measured shortcomings [5].

6.2.7.2.2 At the level of a candidate sequence, BLEU is defined as the geometric mean of precisions of n-grams of increasing size, multiplied by a brevity penalty coefficient:

$$BLEU = BP \cdot \sqrt[n]{\prod_{k=1}^{n} \frac{TP_{n-gram}(k)}{N_{n-gram}(k)}} \tag{2}$$

where $N_{n-gram}(k)$ is the total number of n-grams of size k in the candidate sequence (including duplicates) and $TP_{n-gram}(k)$ is the number of n-grams of size k in the candidate sequence that are present in at least one of the reference sequences.

So it’a solved problem?

8.4.9 Emotion recognition

8.4.9.1 General

The task of emotion recognition is defined in ISO/IEC TR 23281 [4], 7.4.9.

8.4.9.2 Automated metric: accuracy

8.4.9.2.1 General

Accuracy can be applied to evaluate emotion recognition, as specified in 7.2.2.

8.4.9.2.2 Application of the metric

7.2.2.2 applies.

8.4.9.2.3 Implementation of the metric

7.2.2.3 applies.

8.4.9.2.4 Calibration

No widely recognized information on calibration has been identified at the time of the writing.

8.4.9.2.5 Interpretation

No widely recognized information on interpretation has been identified at the time of the writing.

k) Assessment of evaluation reliability: Assessment tools and criteria from data annotation practices can be usefully leveraged to quantify the reliability of the evaluation, by considering the ratings as similar to annotations. This includes for instance measures like the inter-annotator agreement, for which various concrete metrics exist: Krippendorff's alpha and Cohen's kappa which are the most typical ones, but also Fleiss' kappa, Scott's pi, Gwet's AC1, Pearson's correlation coefficient, intraclass correlation coefficient and percent agreement. Measuring low evaluation reliability can lead to revisiting some of the previous steps, such as improving the guidelines or increasing training. This assessment can also be performed in parallel to previous steps, in order to better anticipate issues.

So it’a solved problem?

AI safety Evaluation as a wicked problem

Pluralisitic human values

- Whose values count?
- Which harms matter?
- Which trade-offs are acceptable?

Context dependent

- Users and institutions
- Incentives and deployment conditions
- Behavioral responses to interventions

AI safety evaluation

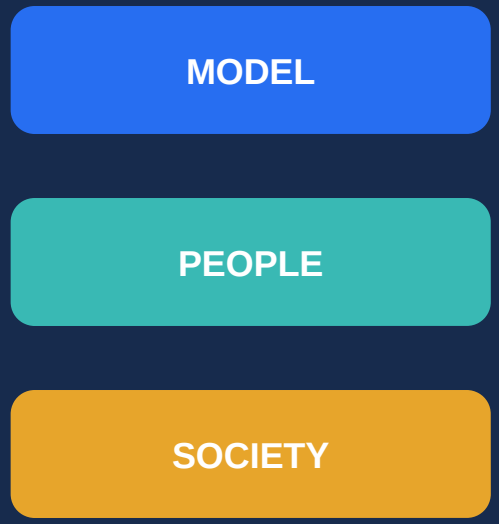
	Model safety / capability	**Contextual safety / impact**
Core question	What outputs can the model produce?	What changes when people or connected systems use it?
Typical focus	Accuracy, reliability, harmful capabilities	Behavior, decisions, task performance
Misinformation example	Can it generate misleading content?	Does that content actually mislead people?

Source: CSET, *AI Safety Evaluations: An Explainer* (2025) [1]

AI SYSTEM EVALUATION

Sociotechnical evaluation

Evaluating the model, its users, and the larger system



Based on Weidinger et al. (2023)

generated with CodeX

MOTIVATION

A benchmark tests only part of the system

Model benchmark

Prompt + model + score

Controlled inputs and predefined metrics

Deployed AI system

Model

Interface and product rules

Users with different goals

Institutions, markets, and norms

A safe component does not guarantee a safe system.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

FRAMEWORK

Three layers of sociotechnical evaluation

1	**Capability** What can the AI system do?	Benchmarks, output analysis, adversarial testing
2	**Human interaction** What happens when people use it?	Behavioral experiments, user research, monitoring
3	**Systemic impact** What emerges when use becomes widespread?	Pilot studies, impact assessments, simulations

The layers add context. Teams can evaluate them in parallel.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

Layer 1: capability evaluation

Target

The model, its data, filters, and outputs

Example: AI news assistant

Test factual accuracy, source attribution, multilingual performance, and adversarial prompts.

What this layer can tell us

- Whether the system can produce harmful outputs
- Which prompts, languages, or groups expose failures
- Whether safeguards reduce or redistribute errors

It indicates possible harm. It does not measure downstream harm by itself.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

Layer 2: human interaction

Same model, different outcomes

Careful reader	Checks sources before sharing
Rushed user	Accepts a fluent answer without checking
Malicious user	Searches for persuasive false claims

Evaluation questions

1. Do users believe incorrect answers?
2. Does the interface create overtrust?
3. Who benefits, and who encounters more failures?
4. Do effects change after repeated use?

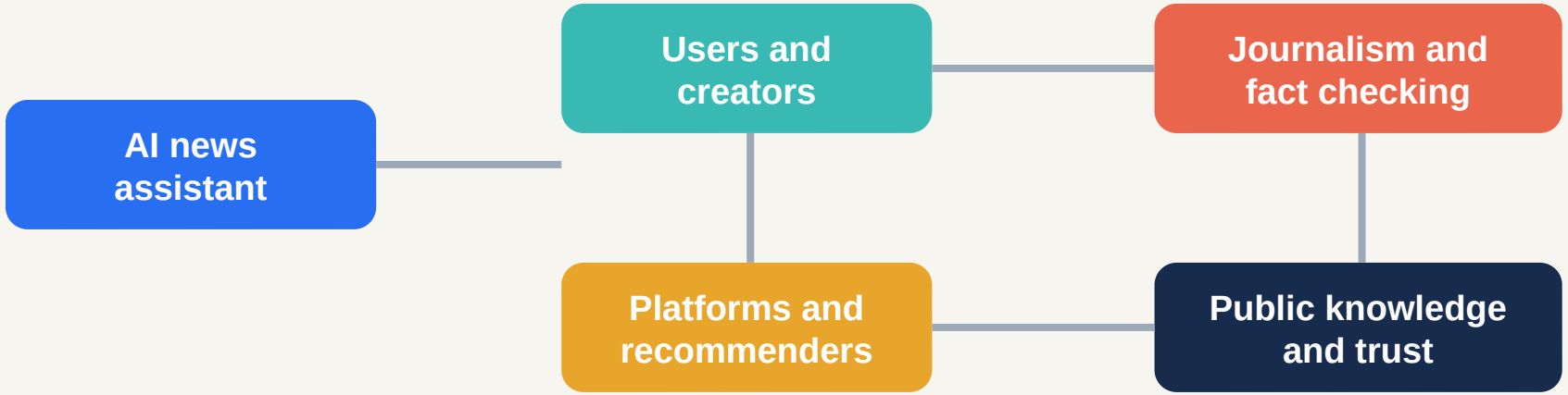
Methods: experiments, user research, passive monitoring, longitudinal studies

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

Systemic impact evaluation

Widespread use creates emergent effects



Measure

Synthetic content prevalence

Trust in shared information

Unequal distribution of benefits and harms

Methods: staged releases, impact assessments, forecasts, and simulations

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

WORKED EXAMPLE

Misinformation across all three layers

CAPABILITY

Can it generate a convincing false story?

Measure factuality, credibility, and robustness.

HUMAN INTERACTION

Do people believe or share the story?

Study persuasion, deception, and overtrust.

SYSTEMIC IMPACT

Does synthetic content alter the information environment?

Track prevalence, public trust, and institutional response.

A complete safety claim needs evidence from every layer.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

A practical evaluation workflow

1	**Risk and affected groups**	Name the possible harm and who may experience it.
2	**Causal pathway**	Explain how model behavior could produce the harm.
3	**Evidence at each layer**	Select technical tests, human studies, and system indicators.
4	**Decision threshold**	State what counts as acceptable and compared with what.
5	**Action and monitoring**	Connect results to design, deployment, restrictions, or withdrawal.

Evaluation choices embed values: what to measure, whose experience counts, and what is safe enough.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

SUMMARY

Sociotechnical evaluation

System outcome = f(model, interface, users, institutions, context)

Capability	What can the system do?
Human interaction	What happens during use?
Systemic impact	What emerges at scale?

Benchmarks remain useful. The larger system determines whether capabilities become harms.

Source: Weidinger et al. (2023), arXiv:2310.11986

generated with CodeX

Mythos System Card

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2.2.1 What we measured

We measured, in several ways, whether the model can provide outputs comparable to a top-tier research team or specialized laboratory.

To do this, we conducted expert red teaming in which experts were asked to compare threat-relevant scientific capabilities of the model to sources and experts of differing caliber. To validate these findings, we also conducted an uplift trial in which we asked biology PhD graduates to construct the same scenarios that experts evaluated the model on, and assessed them for feasibility. Finally, we compared the model's biological sequence-to-function modeling and design capabilities to top performers in the US labor market.

https://www-cdn.anthropic.com/08ab9158070959f88f296514c21b7facce6f52bc.pdf

Relevance	Evaluation	Description
Known and novel CB weapons	**Expert red teaming**	Can models provide uplift in catastrophic chemical/biological weapon development?
Known biological weapons	**Virology protocol uplift trial**	Can models uplift human experts in making a detailed end-to-end protocol for synthesizing a challenging virus?
	Automated medium-horizon evaluations **Long-form virology tasks** **Multimodal virology (VCT)** **DNA Synthesis Screening Evasion**	Can agentic systems complete individual tasks related to acquiring, designing, and synthesizing a virus? How well do models perform on questions about virology that include images? Can models design DNA fragments that bypass gene synthesis screening?
Novel biological weapons	**Catastrophic biological scenario uplift trial**	Can models uplift individuals with PhD training in the construction of scenarios with catastrophic potential?
	Sequence-to-function modeling and design	Can models match expert human performance on a calibrated biological sequence modeling and design task?

[Table 2.2.4.A] Evaluations for chemical and biological weapons.

AI Impact Eval: Beyond Benchmark Scores

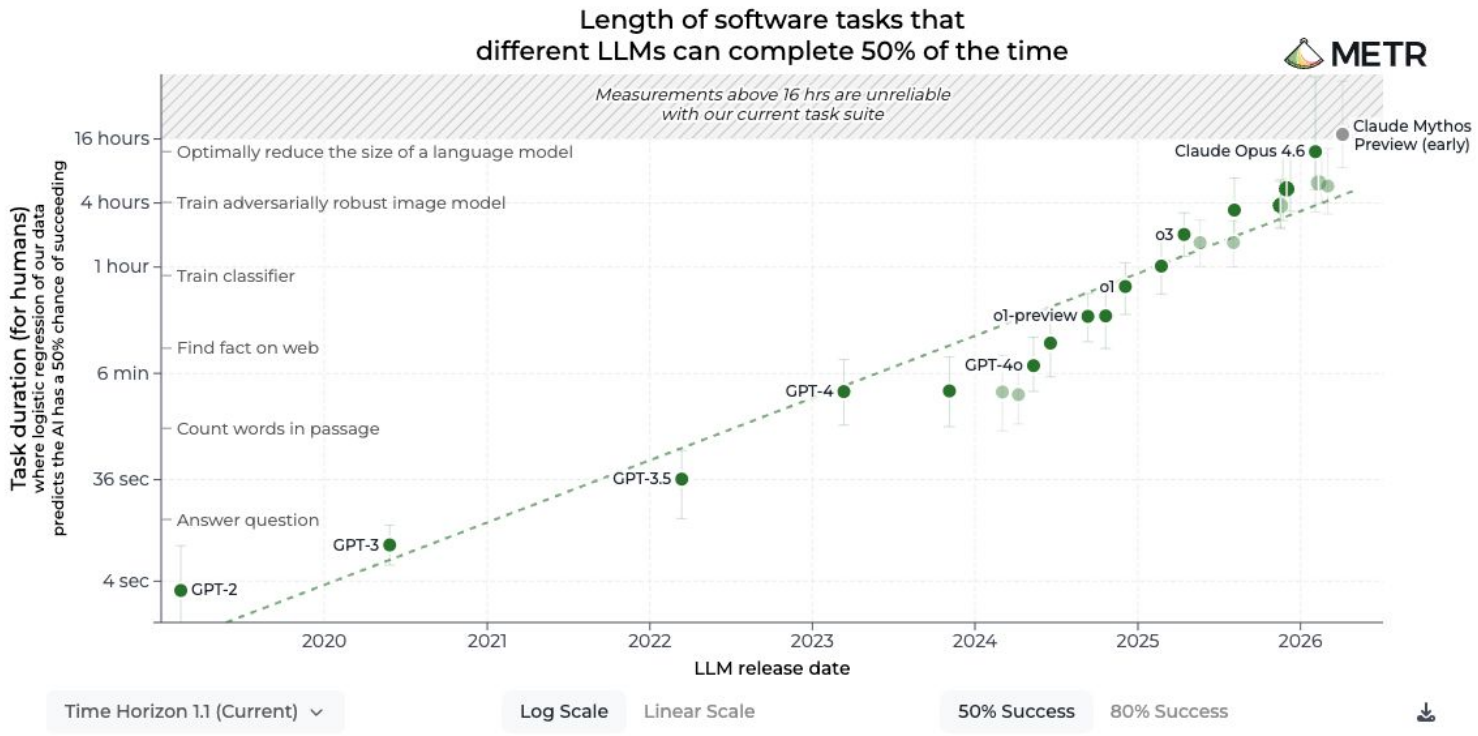
METR’s task-completion time horizon

https://metr.org/time-horizons/

It measures the length of software-related task that an AI agent can complete with a specified probability of success.

For example, a 50% time horizon of two hours means an AI agent is expected to succeed on 50% of the tasks that take a human expert two hours.

It does not mean the AI operates continuously for two hours or can perform every two-hour human task.



Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Table 1: Example tasks for different durations; see Rein et al. [8] and Wijk et al. [2] for more examples.

Kwa, Thomas, et al. "Measuring AI ability to complete long software tasks." *Advances in Neural Information Processing Systems* 38 (2026): 92213-92266.

Pros

- **Interpretability**: Measures capabilities in terms of human work duration rather than abstract scores.
- **Avoids saturation**: Captures continued progress as traditional benchmarks approach ceiling performance.
- **Long-horizon autonomy**: Evaluates complex, multi-step tasks rather than isolated questions.
- **Forecasting**: Enables tracking and extrapolation of AI capability growth.
- **Real-world relevance**: Better reflects potential for task automation and economic impacts.

Cons

- **Limited domain coverage**: Evaluations focus primarily on software engineering and related technical tasks. Findings may not generalize to other domains.
- **Artificially well-defined tasks**: Real-world work often involves ambiguous objectives, interpersonal communication, negotiation, and evolving requirements that are difficult to capture in controlled evaluations.
- **Limited reliability at high-stakes thresholds**: The headline 50% success threshold falls short of the reliability required for many economically valuable or safety-critical tasks. Even the reported 80% horizon may be insufficient.
- **Limited measurement of societal impacts**: The metric measures autonomous task-completion capabilities rather than downstream harms, such as discrimination, labor displacement, or concentration of power.

What is ‘good’ AI

Sources I

1. Ji, Venkatram, and Batalis. AI Safety Evaluations: An Explainer. CSET, 2025.
2. Weidinger et al. Sociotechnical Safety Evaluation of Generative AI Systems. 2023.
3. Regulation (EU) 2024/1689, especially Articles 1, 27, and 55.
4. Council of Europe Framework Convention on AI.
5. ISO/IEC 42005:2025.
6. ISO/IEC 42001:2023.
7. OECD AI Principles.
8. NIST AI RMF Playbook.
9. Council of Europe Convention implementation and HUDERIA.

Sources II

10. Anthropic Responsible Scaling Policy and version history.
11. Anthropic Trust Center: RSP and FCF roles; FCF introduction.
12. Google DeepMind frontier safety.
13. Microsoft Responsible AI Standard introduction.
14. Microsoft Frontier Governance Framework, 2026.
15. OpenAI Preparedness Framework; Frontier Governance Framework.
16. Claude Mythos Preview System Card.
17. Gemini 3.1 Pro Model Card.
18. Microsoft 365 Copilot application card.
19. OpenAI Operator System Card.
20. Anthropic Transparency Hub.

Sources III

21. METR task horizons.
22. Anthropic, Labor market impacts of AI, March 2026.
23. METR early-2025 developer productivity experiment.
24. METR developer productivity study update, February 2026.
25. Brynjolfsson, Li, and Raymond. Generative AI at Work. *Quarterly Journal of Economics*, 2025.

Links and policy versions should be checked before teaching, because living frameworks and system cards can change.